

Ebelechukwu Nwafor

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EDUCATION

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- HOWARD UNIVERSITY, Department of Systems and Computer Science** **Washington, DC**
Doctor of Philosophy in Computer Science *May 2018*
Dissertation Topic: “Trace-based Data Provenance for Cyber-Physical Systems”
Advisor: Dr. Gedare Bloom
- NORFOLK STATE UNIVERSITY, Department of Computer Science** **Norfolk, VA**
Master of Science in Computer Science *May 2013*
Thesis: “Prototype implementation of a self-protecting security framework for CDA documents”
Advisor: Dr. George Hsieh
- CHEYNEY UNIVERSITY OF PENNSYLVANIA,** **Cheyney, PA**
Bachelor of Arts in Computer and Information Science *May 2011*

APPOINTMENTS

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- Villanova University** **Villanova, PA**
- Associate Professor** *Aug 2025- Present*
- Associate Professor where I have taught courses such as Introduction to Cybersecurity, Introduction to Python, Principles of Database Systems, and Security and Privacy in Cyber Physical Systems. In addition to my teaching responsibilities, I conduct research on Internet of Things (IoT) security and contribute to various service activities within the university.
- Assistant Professor** *Aug 2019- Present*
- Tenure-track Assistant Professor where I have taught courses such as Introduction to Cybersecurity, Introduction to Python, Principles of Database Systems, and Security and Privacy in Cyber Physical Systems. In addition to my teaching responsibilities, I conduct research on Internet of Things (IoT) security and contribute to various service activities within the university.
- Howard University** **Washington, DC**
- Instructor** *Aug 2019 - Present*
- Teach Introduction to Data Science courses for the Research Centers in Minority Institutions (RCMI) Data Science Initiative, supporting foundational training in data science.
- Assistant Professor** *Aug 2018- May 2019*
- Instructor for the newly formed Howard University-Google Initiative (also known as Google techexchange). My responsibilities involved teaching Theory of Computation and Intro to Networking courses.
- Embedded System and Security Lab, Howard University** **Washington, DC**
Research Assistant/Project Lead *Jan 2016- July 2018*
- The goal of this project is to provide a framework that collects provenance data on memory constrained cyber-physical systems. This information is used to provide anomaly detection on memory contained cyber-physical systems.

PUBLICATIONS

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- Nwafor, E., Olufowobi, H. Towards an Interactive Visualization Framework for IoT Device Data Flow. in the Proceedings of the Second International Workshop on the Internet of Things Data Analytics (IoTDA), Los Angeles, California. December 2019
 - Nwafor, E., Anietie, A. “A Survey on Machine Translation work on Nigerian Languages.” in the Proceedings of the 13th International Conference on Language Resources and Evaluation (LREC 2022).
 - Nwafor, E., Robson, M., Olufowobi, H. Dynamic Load Sharing for Memory Constrained devices: a survey. 2021 IEEE 7th World Forum on Internet of Thing (WF-IoT). 2021

- Nwafor, E., Su, S. "Detecting Network Traffic Intrusions on Memory Constrained Embedded Systems." In the Proceedings of the 2021 Virtual IEEE International Symposium on Technologies for Homeland Security.
- Nwafor, E., Kolimago, C., Vaughn, R., Covid Vaccine Sentiment Analysis by Geographic Regions. In the Proceedings of the IEEE Big Data Workshop for COVID-19.
- Nwafor, E., Olufowobi, H., CANBERT: A Language-based Intrusion Detection Model for In-vehicle Networks. In 2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA), 2022.
- Agbaje, P., Anjum, A., Talukder, Z., Islam, M., Nwafor, E., Olufowobi, H. FedCime: An Efficient Federated Learning Approach For Clients in Mobile Edge Computing. 2023 IEEE International Conference on Edge Computing and Communications (EDGE). 2023.
- Agbaje, P., Nwafor, E., Olufowobi, H. Deep Reinforcement Learning for Energy-Efficient Task Offloading in Cooperative Vehicular Edge Networks. 2023 IEEE 21st International Conference on Industrial Informatics (INDIN). 2023
- Nwafor, E., Schmidt, C., Olufowobi, H., IoT-MGSec: Mitigating Man-in-the-Middle Attacks in IoT Networks using Graph-Based Learning. In 2023 22nd IEEE International Conference on Machine Learning and Applications (ICMLA), 2023.
- Agbaje, P., Anjum, A., Nwafor, E., Olufowobi, H. Privacy-Preserving Intrusion Detection System for Internet of Vehicles using Split Learning. The 10th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT). 2023.
- Agbaje, P., Anjum, A., Nwafor, E., Olufowobi, H. Towards named data networking technology: Emerging applications, use cases, and challenges for secure data communication. Future Generation Computer Systems. 2024
- Landry, L., Andy, A., Lucas, M., Nwafor, E. Artificial Intelligence Assisted Curation of Population Groups in Biomedical Literature. 18th International Digital Curation Conference (DCC). 2024.
- Nwafor, E., Robson, M., Olufowobi, H. Simulating Load Sharing for Resource Constrained Devices. IEEE Access Journal (Under Review). 2024.
- Nwafor, E., Ujjwal, B., Parwez, S., Blackstone, J., Olufowobi, H. Evaluating Large Language Models for Enhanced Intrusion Detection in Internet of Things Networks. The 2024 IEEE Global Communications Conference (GLOBECOM). 2024.